AYE
TECH HUB

• PLC PROGRAMMING • LESSON 02

PLC Hardware Deep Dive

CPU, I/O Modules, Addressing & Field Wiring

Lesson 2 of 30 | PLC Programming Program | Beginner

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- CPU architecture — processor, memory types, I/O image table
- Digital and analog I/O modules — specs, wiring, and signal types
- Sinking vs sourcing I/O — NPN/PNP sensors explained
- I/O addressing — how Siemens, Allen-Bradley, and Mitsubishi number their I/O
- Power supply, grounding, and panel wiring best practices

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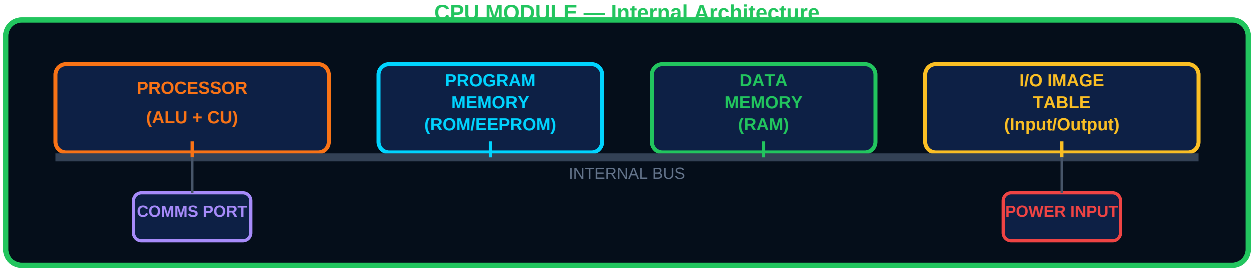
01 | The PLC CPU — Brain of the Controller

The **CPU (Central Processing Unit)** module is the intelligence of the PLC. It contains the processor that runs the control program, multiple types of memory for storing the program and runtime data, and the communications interfaces for talking to HMIs, SCADA systems, and other PLCs. Understanding what happens inside the CPU helps you write better programs and diagnose faults faster.

CPU MODULE — Internal Architecture



Memory Type	Full Name	Contents	Volatile?	Typical Size
ROM / Flash	Read-Only Memory / Flash EEPROM	Operating system firmware, CPU boot code, fixed factory settings	No — survives power loss	1–16 MB
EEPROM / Load Memory	Electrically Erasable Programmable ROM	User PLC program, data blocks, hardware configuration — what you download from TIA Portal / Studio 5000	No — survives power loss	2–256 MB (card)
RAM / Work Memory	Random Access Memory	Active running copy of program, I/O image table, timer/counter values, data tags in execution	YES — lost on power loss (battery backup restores)	512 KB – 8 MB
Retentive RAM	Battery-backed RAM	Retain values through power cycles: recipes, production counts, setpoints	No — battery keeps data alive	Subset of RAM
I/O Image Table	Process Image (Siemens) / I/O Data Table (AB)	Snapshot of all input states (read at scan start) and pending output values (written at scan end)	YES — refreshed every scan	Bytes matching I/O count

02 | The I/O Image Table — Why Inputs Are Frozen Mid-Scan

At the start of every scan cycle, the CPU reads all physical input terminals and copies their states into a **memory area called the I/O Image Table** (or Process Image in Siemens). The program then reads from this memory copy — not directly from the physical terminals — for the entire scan. This means even if a sensor changes state mid-scan, the program sees the value it had at scan start. At scan end, output image values are written to the physical output terminals.

WHY THIS MATTERS — Deterministic and Consistent Execution

Because the program reads a frozen snapshot of inputs, all rungs in the program see a consistent world — the same input values throughout the entire scan. This makes PLC programs deterministic and predictable. If inputs were read live mid-program, one rung could see a button pressed while another sees it released in the same scan — causing unpredictable and dangerous machine behaviour.

03 | I/O Modules — Digital and Analog in Detail

Module Type	Signal Type	Common Voltage Ranges	Typical Field Devices	Key Spec
Digital Input (DI)	ON/OFF discrete	5V, 24V DC, 110V AC, 240V AC	Pushbuttons, limit switches, proximity sensors, floats, pressure switches	No. of channels (8/16/32), response time (ms), input impedance
Digital Output (DO) Transistor (DC)	ON/OFF DC switching	24V DC typical	Solenoids, small DC motors, indicator lamps, relay coil inputs	Current rating (0.5–2A/ch), short-circuit protection, response time (< 1ms)
Digital Output (DO) Relay	ON/OFF — volt-free contacts	Any — relay switches customer power	Motors (via starters), AC solenoids, heaters, alarms, devices of any voltage	Contact rating (2–10A), voltage range, lifetime (mechanical / electrical cycles)
Digital Output (DO) Triac (AC)	ON/OFF AC switching	110/240V AC	AC heaters, AC solenoids, lighting, small AC motors	Current rating, zero-crossing switching, heat dissipation
Analog Input (AI)	4–20 mA or 0–10V DC	Standard: 4–20 mA (current loop)	Temperature transmitters (RTD/TC via transmitter), pressure, flow, level sensors	Resolution (12 or 16-bit), channel count, input filter time constant
Analog Output (AO)	4–20 mA or 0–10V DC	Standard: 4–20 mA	Variable speed drives (VFDs), control valves, proportional solenoids, pumps	Resolution, settling time, load resistance (max 600Ω typical for mA output)

04 | Sinking vs Sourcing — NPN and PNP Sensors

One of the most confusing topics for beginners is the difference between **sinking (NPN)** and **sourcing (PNP)** I/O. The terms refer to the direction of current flow. The sensor type must match the input module wiring type — a mismatch means the input will never work, no matter how correct your ladder logic is.

Type	Current Direction	Sensor Output Wire	Connects To	Common In
NPN (Sinking)	Current flows INTO the sensor output — sensor sinks current to 0V	Output pulls to 0V (GND) when active	Sinking input module — common (COM) connected to +24V	Japan (Mitsubishi, Omron, Keyence)
PNP (Sourcing)	Current flows OUT OF the sensor output — sensor sources current from +V	Output pulls to +24V when active	Sourcing input module — common (COM) connected to 0V	Europe (Siemens, Schneider, Sick)
Universal (NPN+PNP)	Module accepts both types simultaneously	Automatic detection	Any sensor of either type	Modern Siemens S7-1200/1500, Allen-Bradley Micro850

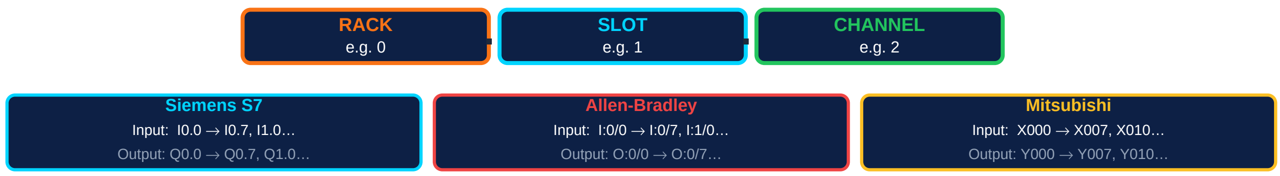
QUICK CHECK — Which Type Is Your Sensor?

> Look at the sensor datasheet or label. NPN sensors are marked "NPN", "Sinking", or show the output switching to 0V. PNP sensors are marked "PNP", "Sourcing", or show the output switching to +VCC. In Europe, PNP is the default. In Japan, NPN is the default. Most modern industrial proximity sensors come in both versions — order the correct type for your PLC input module.

05 | I/O Addressing — How PLCs Name Their Channels

Every input and output channel in a PLC has a unique address that the program uses to reference it. The addressing scheme varies between manufacturers but always encodes the same information: which rack, which slot (module), and which channel within that module.

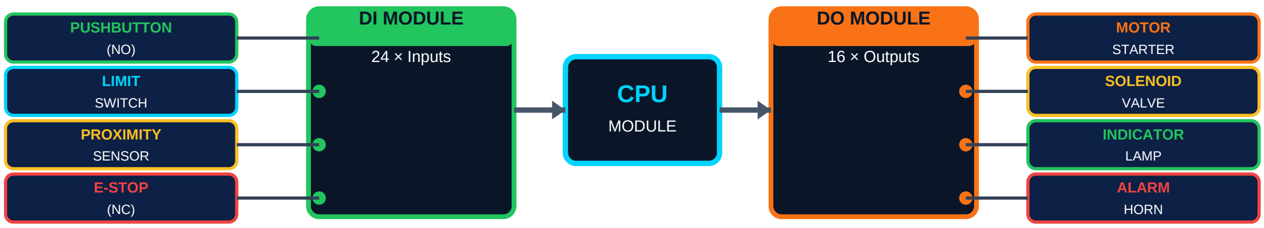
I/O Address Structure — How PLCs Reference Field Devices



Brand	Input Format	Output Format	Example Read	Example Write	Bit vs Word
Siemens S7 (IEC)	I[Byte].[Bit]	Q[Byte].[Bit]	I0.0 = DI slot 0, ch 0	Q0.3 = DO slot 0, ch 3	IW0 = 16-bit input word 0
Allen-Bradley (SLC)	I:[Slot]/[Bit]	O:[Slot]/[Bit]	I:1/0 = Slot 1, bit 0	O:2/7 = Slot 2, bit 7	I:1.0 = 16-bit word
Allen-Bradley (Logix)	LocalIN:[Slot].[Ch]	LocalOUT:[Slot].[Ch]	Local:1:I.Data.0	Local:2:O.Data.3	Tag-based — named tags, not numbers
Mitsubishi (FX)	X[Octal]	Y[Octal]	X000, X001…X007, X010	Y000, Y001…Y007, Y010	Octal numbering — no 8 or 9
Omron (CJ/CP)	CIO [Word].[Bit]	CIO [Word].[Bit]	CIO 0.00 (Word 0, bit 0)	CIO 100.00 (output area)	CIO area 0–99 usually inputs

06 | Power Supply & Panel Wiring Best Practices

Field Device Wiring — Inputs (Left) → PLC → Outputs (Right)



Wiring Rule	Requirement	Why It Matters
Dedicated 24V DC supply	Use a DIN rail SMPS (Switched Mode Power Supply) — never share with non-PLC loads	Voltage spikes from contactors/motors can corrupt PLC memory or damage I/O modules
Separate AC and DC wiring	Run AC power cables (230V) on left side of duct; DC signal wires on right side	AC cables induce noise into DC signal wires — causes random I/O toggling

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Earth/ground all metalwork	Bond the control panel enclosure, DIN rail, and cable shielding to PE (Protective Earth)	Static and induced noise is diverted to earth rather than corrupting signal wires
Ferrite cores on sensor cables	Add ferrite beads on sensor cable entry into panel at VFD installations	VFDs radiate high-frequency noise onto signal cables — ferrites block it
Fuse each output	Each DO relay output should have its own fuse sized for the load	A short-circuit actuator blows only that fuse — not the entire output module
Label every wire	Use wire ferrules and numbered sleeves matching the panel drawings	Unlabelled wires are unserviceable — commissioning and fault-finding can take 10× longer

Reflection:

> *"Look at any control panel near you — a machine, a lift, a pump, or a distribution board. Identify the power supply, the PLC rack, the input module terminal strip, and the output module terminal strip. Trace one wire from a field device (sensor or button) to its input terminal. What address would that channel have in Siemens notation? In Allen-Bradley notation?"*

Action checklist:

Next up — PLC-003:
I/O Wiring and Addressing — hands-on wiring exercises for sensors, solenoids, and motor starters to real PLC terminals.

Community & support:
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